

Tomas Busa

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Education

Imperial College London – MSc in Advanced Computing Sep 2026

Relevant Coursework: Systems Performance Engineering, Advanced Computer Security, Networked Systems

Publication: *EPS: Programmable System calls are all you need*, **Tomáš Buša** et al. – *under review*, EuroSys '27

Vrije Universiteit Amsterdam – BSc in Computer Science Aug 2025

GPA: 8.8, Cum Laude, Honours Programme, GLOBE Scholarship

Hong Kong University of Science and Technology – Exchange, Computer Science and Engineering Dec 2024

Relevant Coursework: Discrete Math, Calculus, Data Mining, Advanced Construction using AI and Robotics

Experience

Software Engineer Intern, MicroStep – Bratislava, Slovakia June 2024 – Aug 2024

- Engineered an automation tool to parse electrical schematics of CNC machines and generate YAML, EBI, and ENI configuration files for EtherCAT devices, eliminating manual creation and reducing setup time by 80%
- Developed generators for I/O change scripts to automate I/O map updates, significantly reducing manual intervention and error rates, thus enabling faster integration of new hardware

Software Engineer Intern, MicroStep – Bratislava, Slovakia June 2023 – Aug 2023

- Rebuilt a socket-based application, enabling real-time communication between the CNC control PC and the MMI PC, reducing reliance on internal COBs and lowering latency by 30%
- Remodelled and built a regex-based file parsing library, used to process configuration and log files across CNC automation tools, enabling code reuse and reducing code duplication
- Upgraded a Python-based C code generator by integrating a parsing library, used for generating I/O mapping and device driver code in CNC systems, which reduced manual edits by 30%

Data Analyst Intern, MicroStep – Bratislava, Slovakia June 2022 – Aug 2022

- Analysed and visualised financial and operational data using Power BI, to evaluate CNC machine production cash flow, supporting data-driven strategic purchasing decisions

Projects

Accelerated Networking (AccNet) – Transparent Kernel-Bypass Networking

- Built AccNet's 2,350-line eBPF interception layer on EPS (eBPF-Programmable system calls) enter/exit hooks: 21 socket system calls of unmodified TCP applications are rewritten or implemented in eBPF
- Built the data path on per-connection eBPF ring buffers (RINGBUF for TX, USER_RINGBUF for RX) shared with Machnet, Microsoft's DPDK stack
- Co-developed bpf-next kernel patches: eventfd kfuncs that keep socket readiness correct from eBPF, and writable ring buffer mappings for in-place packet headers and zero-copy DMA
- Handled fork and crash cleanup with CO-RE tp_btf programs that sweep per-process state
- Versus tuned Linux TCP: 1.59–2.59× lower median latency, ~2× throughput for unmodified memcached and lighttpd, and 2.34× for a Go server that libc-level interception cannot reach
- Source code private pending paper review (EuroSys '27)

Spectre-PHT Detection Tool github.com/YBush1/angr-speculative-pht

- Modelled and developed a binary-level Spectre-PHT detector by extending the angr symbolic execution engine to model speculative execution, providing an approach that does not require relational analysis
- Benchmarked on standard Spectre-PHT litmus gadgets, achieving 100% precision/recall

Skills

Languages: Python, C, C++, Java

Technologies: DPDK, Shared-memory IPC, Git, Linux Programming, Bash, Docker, Kubernetes, Azure, REST API, PostgreSQL, Socket Programming, GitHub Actions, Unit Testing, angr, Static Analysis, gdb

eBPF: libbpf, CO-RE, ring buffers / user ring buffers, map-in-map, tp_btf, bpftool, verifier debugging